Power Gated SRAM Circuits with Data Retention Capability and High Immunity to Noise: A Comparison for Reliability in Low Leakage Sleep Mode

Hailong Jiao and Volkan Kursun

Department of Electronic and Computer Engineering
The Hong Kong University of Science and Technology
Clear Water Bay, Kowloon, Hong Kong
{jiaohl, eekursun}@ust.hk

Abstract — **A new power gated 6T SRAM circuit is proposed in this paper to suppress leakage power consumption in data retention SLEEP mode. A new write assist circuitry is presented to enhance the write margin of the new power gated memory circuit. Design tradeoffs among data stability, power consumption, and write margin are evaluated with different SRAM circuits. The leakage power consumption is reduced by up to 3.84x and the read static noise margin is increased by up to 4.79x with the new memory power gating technique as compared to a previously published power gated 6T SRAM circuit in a UMC 80nm CMOS technology.**

Keywords — Power gating, cache, data stability, leakage power consumption, single-ended write, write assist circuitry.

I. INTRODUCTION

With continuous technology scaling, leakage currents dominate the total energy consumption of integrated circuits [1]. Memory banks are primary producers of leakage currents in high performance microprocessors and systems-on-chips (SoCs) [2]. As increasing numbers of SRAM cells are crammed onto integrated circuits to meet the aggressive performance requirements, the leakage power consumed by memory arrays increases with each technology generation. Data stability is another important concern associated with SRAM cells. With reduced supply voltage, smaller devices, and exacerbated process variations, maintaining data stability has become a primary challenge in SRAM design. New memory circuits with lower leakage and higher data stability characteristics are therefore highly desirable.

Power gating is the most popular technique for suppressing leakage power consumption in idle integrated circuits [1] – [5]. The data in memory cells are however lost when standard power gating techniques are applied to an SRAM array. Specialized power gating techniques are required to maintain the data while reducing the leakage power consumption in idle memory circuits.

In this paper, a novel power gating technique is proposed. Data is maintained in the power gated SRAM cell with high data stability in a low-leakage SLEEP mode. A new write assist circuitry is also presented to provide a sufficiently high write margin with the new SRAM circuit. The tradeoffs among various important design metrics are evaluated with different memory circuits.

The paper is organized as follows. A previously published power gated SRAM circuit is reviewed in Section II. The novel power gated 6T SRAM circuit is presented in Section III. The data stability, leakage power, active power, and write margin of different 6T memory circuits are characterized in Section IV. The paper is concluded in Section V.

II. OLD POWER GATED 6T SRAM CIRCUIT

A power gating technique is presented in [4] to suppress the leakage currents while maintaining the data in idle SRAM circuits. This previously published (old) power gated 6T SRAM circuit designed with a UMC 80nm multi-threshold voltage CMOS technology [10] is shown in Fig. 1. Standard performance high-$|V_{th}|$ (SPHVT), standard performance regular-$|V_{th}|$ (SP), and standard performance low-$|V_{th}|$ (SPLVT) transistors are provided with the UMC 80nm CMOS technology that is used in this study as listed in Table I.

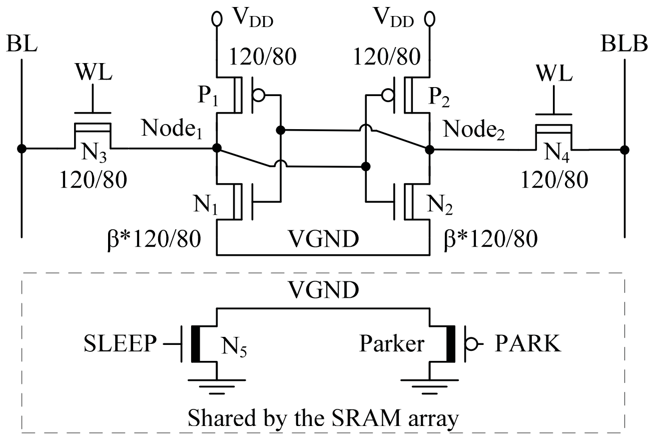

Fig. 1. Old power gated 6T SRAM circuit (OldPG) [4]. High-$|V_{th}|$ transistors (SPHVT) are represented with a thick line in the channel region. Regular-$|V_{th}|$ transistors (SP) are represented with double lines in the channel region. The transistor sizes (Width/Length) are in nanometers assuming an 80nm CMOS technology. WL: wordline. BL: bitline. BLB: bitline bar. VGND: virtual ground line.

TABLE I. UMC 80nm CMOS TECHNOLOGY

	SPHVT	SP	SPLVT
V_{DD} (V)	1	1	1
NMOS V_{th} (mV)	370	240	155
PMOS V_{th} (mV)	–310	–177	–105

In a conventional 6T SRAM cell as shown in Fig. 1, the stored data is disturbed due to the voltage division between the cross-coupled inverters and the bitline access transistors during a read operation [5] – [8]. The widths of the pull down transistors in cross-coupled inverters are typically tuned to tradeoff data stability, data access speed, cell area, and leakage power consumption in conventional 6T SRAM cells. The data is however vulnerable to external noise even with a large β ratio (see Fig. 1) due to the direct read access mechanism of a conventional 6T SRAM cell as discussed in [5] – [8]. The 6T SRAM cells in the old power gated memory array are composed of only SP transistors to provide sufficient data stability as illustrated in Fig. 1.

In the ACTIVE mode, the centralized high-$|V_{th}|$ footer (N_5) in Fig. 1 is turned on while the high-$|V_{th}|$ PMOS Parker is cut off. The voltage on the virtual ground line (VGND) is maintained at ~V_{gnd}. The SRAM circuit therefore operates with high performance. In the SLEEP mode, N_5 is cut off while the Parker is turned on. The voltage of the virtual ground line rises to the threshold voltage of the Parker ($|V_{tp}|$). A reduced yet significant voltage difference ($V_{DD} - |V_{tp}|$) is experienced by the cross-coupled inverters in the SRAM cells. The data is thereby maintained while the leakage currents produced by the idle memory array are reduced.

III. A Novel Power Gated 6T SRAM Circuit

The novel power gated 6T SRAM circuit is presented in this section. The primary goals of the new circuit technique are to reduce leakage power consumption and enhance data stability in idle memory banks. The schematic of the new 6T SRAM circuit is shown in Fig. 2. The six transistors (P_1, N_1, P_2, N_2, N_3, and N_4) within the SRAM cell are all minimum sized to reduce area overhead as illustrated in Fig. 2. The data is stored in the high-$|V_{th}|$ cross-coupled inverters (formed by P_1, N_1, P_2, and N_2). The low-$|V_{th}|$ N_{RA} is part of the read port and shared by all the cells in the same row of the SRAM array. A centralized high-$|V_{th}|$ NMOS footer sleep transistor (N_5) is connected to the sources of all read assist transistors (N_{RA}) in the SRAM array to suppress read bitline leakage. Furthermore, a write assist circuitry composed of N_{CH} and N_{WR} is connected to the source of N_1 to enhance the write ability of the SRAM cell as shown in Fig. 2. The write assist circuitry and the cell virtual ground line (C_VGND) are shared by all the cells of a memory word.

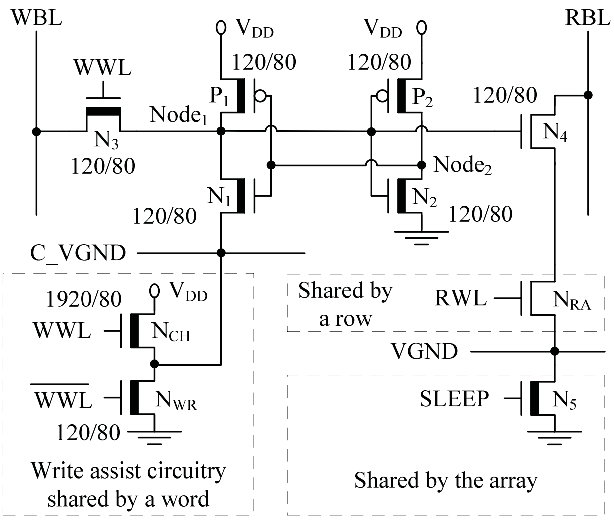

Fig. 2. The new power gated 6T SRAM circuit (NewPG). High-$|V_{th}|$ transistors (SPHVT) are represented with a thick line in the channel region. Low-$|V_{th}|$ transistors (SPLVT) are represented with a thin line in the channel region. The transistor sizes (Width/Length) are in nanometers assuming an 80nm CMOS technology. WBL: write bitline. RBL: read bitline. WWL: write wordline. RWL: read wordline. C_VGND: cell virtual ground line. VGND: array virtual ground line.

Prior to a write operation, the write bitline WBL is charged to V_{DD} or discharged to V_{gnd} depending on whether a "1" or a "0" is to be written onto $Node_1$. The write wordline WWL transitions high to initiate a single-ended write operation. The read wordline RWL and SLEEP are maintained at V_{gnd} and V_{DD}, respectively. The data is forced from WBL onto $Node_1$ through N_3. In order to successfully force a "0" onto $Node_1$, N_3 must be stronger as compared to P_1. Alternatively, in order to successfully force a "1" onto $Node_1$, N_3 must be stronger as compared to N_1. However, both N_1 and N_3 are minimum sized to reduce the area overhead as shown in Fig. 2. Furthermore, there is a threshold voltage drop across N_3 when a "1" is transmitted from WBL onto $Node_1$. A write assist circuitry is employed in this study to enhance the write margin while transferring a "1" into the new SRAM cell. Since WWL is high during the write operation, N_{WR} is cut off and N_{CH} is turned on. C_VGND is charged by N_{CH} to a voltage level significantly higher than V_{gnd}. N_1 is weakened by increasing the C_VGND voltage, thereby enhancing the write margin while transferring a "1" into the SRAM cell. Furthermore, $Node_1$ is charged by the current produced by N_{CH} and N_1, thereby facilitating the transfer of a "1" into the SRAM cell and further increasing the write margin. When the write operation is finished, WWL transitions low. N_{CH} is cut off and N_{WR} is turned on. C_VGND is discharged to ~V_{gnd} by N_{WR}. After the write operation, WBL does not need to be recharged to V_{DD}. The write operation can therefore be extended to the end of a clock cycle with the new SRAM circuit, thereby accommodating a longer write access delay while maintaining a high clock frequency.

Prior to a read operation, RBL is charged to V_{DD}. RWL transitions to V_{DD} to initiate a single-ended read operation. WWL and SLEEP are maintained at V_{gnd} and V_{DD}, respectively. Provided that a "1" is stored at $Node_1$, RBL is discharged through the read port composed of N_4, N_{RA}, and N_5. Alternatively, RBL is maintained at V_{DD} if a "0" is stored at $Node_1$. Since the data is not directly accessed during a read operation, the read static noise margin (read SNM) is significantly enhanced as compared to the conventional 6T SRAM cell. Furthermore, in the write assist circuitry, N_{WR} is turned on and N_{CH} is cut off during a read operation. The voltage of C_VGND is maintained at ~V_{gnd}. The read SNM is thereby not degraded with the write assist circuitry. When the read operation is finished, RBL is recharged to V_{DD} to prepare for the next read operation.

In the SLEEP mode, WWL, RWL, and SLEEP are all maintained at ~V_{gnd}. The high-$|V_{th}|$ N_3 and N_5 are cut off. The write and read bitline leakage currents are thereby significantly suppressed. Furthermore, since the cross-coupled inverters are composed of high-$|V_{th}|$ transistors as shown in Fig. 2, the cell leakage is minimized. No sneak leakage current path is introduced by the write assist circuitry since N_{CH} is cut off in the SLEEP mode. Alternatively, since N_{WR} is turned on, C_VGND is maintained at ~V_{gnd}. The cross-coupled inverters experience full rail-to-rail supply voltage. The NewPG memory cells thereby provide high data hold stability in the low leakage SLEEP mode.

IV. Characterization of SRAM Circuit Techniques

The UMC 80nm multi-threshold voltage CMOS technology is used in this paper for the characterization of different SRAM circuits. 8Kb SRAM arrays are designed based on the following techniques: standard single regular-$|V_{th}|$ 6T SRAM circuit without power gating (STANDARD), old power gated 6T SRAM circuit (OldPG) shown in Fig. 1 [4], and the new power gated 6T SRAM circuit (NewPG) shown in Fig. 2. Regular-$|V_{th}|$ transistors are used for the peripheral circuitry of the standard single regular-$|V_{th}|$ SRAM circuit. Alternatively, low-$|V_{th}|$ transistors are used for the peripheral circuitry of power gated SRAM circuits (OldPG and NewPG). Power gating is used extensively for the low-$|V_{th}|$ peripheral circuitry. Different design metrics of SRAM circuits are evaluated in this section with HSPICE simulation. The simulation temperature is 90˚C [9].

The footer sleep transistors (N_5 in Figs. 1 and 2) of the power gated SRAM arrays are sized to achieve less than 10mV resistive voltage drop on the virtual ground line (VGND) [5]. The Parker in the OldPG SRAM array is sized to equalize the hold SNM (in the SLEEP mode) and read SNM of the OldPG SRAM cell. N_{RA} in the NewPG SRAM circuit is sized to achieve similar (within 1%) read access time as compared to the OldPG SRAM circuit. The sizes of footers, Parkers, and N_{RA} for different β ratios (the value of β in the following tables and figures of the paper is the β ratio of conventional 6T SRAM cells shown in Fig. 1) are listed in Table II.

Section IV is organized as follows. The data stability of SRAM cells are evaluated in Section IV-A. The leakage power consumed by different SRAM arrays in the SLEEP mode is compared in Section IV-B. The active power consumption with the SRAM circuits is presented in Section IV-C. The write margins of SRAM cells are characterized in Section IV-D.

TABLE II. TRANSISTOR SIZES (μm) IN POWER GATED SRAM ARRAYS

	β = 1	β = 2	β = 3
N_5 (OldPG)	206	258	283
Parker (OldPG)	0.40	0.86	1.46
N_5 (NewPG)	444	513	534
N_{RA} (NewPG)	16.0	20.8	22.6

A. *Data Stability*

The data in conventional 6T SRAM cells (used in the STANDARD and OldPG memory arrays) are most vulnerable to external noise during a read operation. The read static noise margin (SNM) is determined by the voltage transfer characteristics (VTC) of the cross-coupled inverters in each SRAM cell. Read static noise margins of the SRAM cells are compared in Fig. 3.

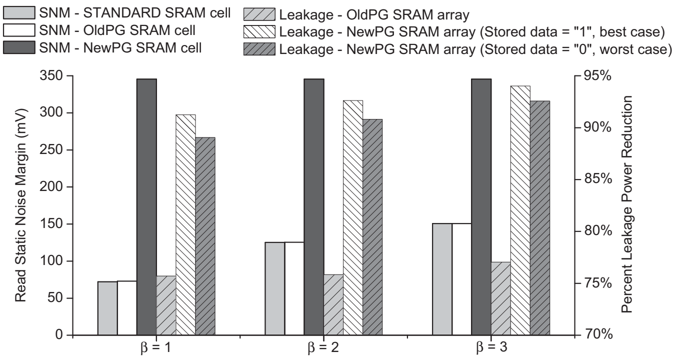

Fig. 3. The read static noise margins (SNMs) of different SRAM cells. The percent leakage power reductions (provided with power gated SRAM arrays in the SLEEP mode as compared to the single regular-$|V_{th}|$ 6T SRAM array) are also shown.

As discussed in Section III, the data in the NewPG SRAM cell is isolated from the read bitline. The data is not disturbed during a read operation. The VTC of cross-coupled inverters in the NewPG SRAM cell are therefore maintained symmetric during a read operation. Alternatively, the VTC of cross-coupled inverters in the conventional 6T SRAM cells are asymmetric due to the voltage rise at the node which stores a "0" during a read operation. The data stability of conventional 6T memory cells is therefore significantly degraded [5] – [8]. Furthermore, the high-$|V_{th}|$ cross-coupled inverters in the NewPG SRAM cell have narrower transition regions, thereby enhancing the data stability as compared to the regular-$|V_{th}|$ inverters in the conventional 6T SRAM cell [3]. As shown in Fig. 3, the read SNM is increased by 2.29x to 4.79x with the NewPG SRAM cell as compared to the conventional 6T SRAM cells depending on the β ratio.

B. *Leakage Power Consumption*

The leakage power consumption of SRAM arrays in the SLEEP mode is evaluated in this section. The percent leakage power reduction with power gated SRAM arrays as compared to the STANDARD SRAM array (a circuit with standard performance regular-$|V_{th}|$ transistors and without any power gating) is shown in Fig. 3.

In the NewPG SRAM circuit, the high-$|V_{th}|$ footer and write access transistors (N_5 and N_3 in Fig. 2) reduce the bitline leakage currents significantly. Furthermore, the high-$|V_{th}|$ cross-coupled inverters in the NewPG SRAM cell effectively suppress the cell leakage currents. The NewPG SRAM array therefore achieves the lowest leakage power among the SRAM arrays evaluated in this paper. The NewPG SRAM array reduces the leakage power consumption by up to 94.02% and 73.97% as compared to the STANDARD and OldPG SRAM arrays, respectively.

The footer is cut off while the Parker is turned on with the OldPG SRAM array in the SLEEP mode. The leakage currents are suppressed due to the increased virtual ground line voltage (lower effective supply voltage and $|V_{DS}|$) as compared to the STANDARD SRAM array. Leakage currents produced by the SP memory cells however freely flow through the active Parker in the OldPG memory array. The leakage power consumption of the OldPG memory circuit is therefore up to 3.84x higher as compared to the NewPG SRAM array.

C. *Active Power Consumption*

The active power consumption of memory arrays are characterized in this section. The write and read power consumptions of SRAM circuits are compared in Fig. 4.

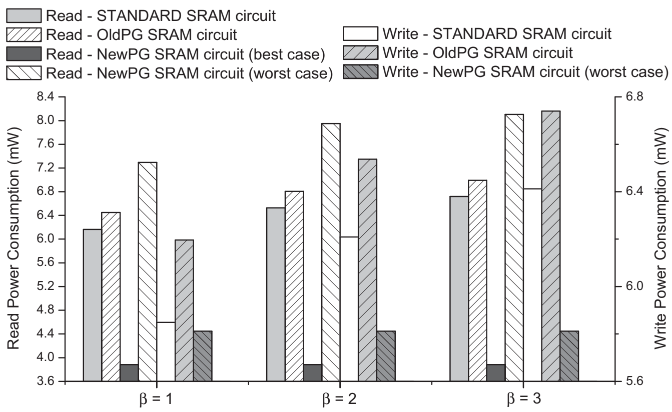

Fig. 4. Read and write power consumptions of different SRAM circuits.

In the NewPG SRAM array, provided that all the words in the same row share a write wordline, an unintentional write operation is initiated in 7 out of 8 words of a row that is accessed for a write operation. In order to avoid the unintentional loss of data in unselected memory words, separate write wordlines are employed in this study for different words in the same row. Separation of write wordlines also eliminates the unnecessary switching activity on the write bitlines of unselected cells. The (worst case) write power consumption of the NewPG SRAM circuit is thereby reduced by up to 13.77% and 9.36% as compared to the OldPG and STANDARD SRAM circuits, respectively, as shown in Fig. 4.

During a read operation, no switching activity occurs on any read bitlines in the NewPG SRAM array when "0" is stored in all the cells. Up to 44.49% and 42.23% read power savings are thereby achieved with the NewPG SRAM circuit (when "0" is stored in all the cells, best case) as compared to the OldPG and STANDARD SRAM circuits, respectively. Alternatively, when "1" is stored in all the cells, the read power consumption (worst case) is increased by up to 17.89% and 14.38% with the NewPG SRAM circuit as compared to the STANDARD and OldPG SRAM circuits, respectively.

D. Write Margin

Write margin [6] is the metric used to characterize the write ability of an SRAM cell. The write margins of different SRAM cells are listed in Table III.

TABLE III. WRITE MARGINS (mV) OF SRAM CELLS

		$\beta = 1$	$\beta = 2$	$\beta = 3$
STANDARD		547	472	435
OldPG		547	475	438
NewPG	Write "0"	296		
	Write "1"	362 (worst case) ~ 579 (best case)		

The write margin for transferring a "1" into the NewPG SRAM cell is strongly dependent on the existing (pre-write) data in the 16-bit memory word. As listed in Table III, while transferring a "1" into the NewPG SRAM cell, the write margin varies between 362mV and 579mV depending on the existing (pre-write) data in the 16-bit memory word. The worst case write margin while transferring a "1" into the NewPG SRAM cell is 16.78% lower as compared to the conventional 6T SRAM cells with $\beta = 3$.

Due to the single-ended write operation with the NewPG SRAM cell, write margins for transferring a "0" and a "1" are different. The write margin while transferring a "1" into the NewPG SRAM cell is up to 1.96x wider as compared to the transfer of a "0". Since no specialized write assist circuitry is employed to enhance the ability to write "0", the write margin while transferring a "0" into the NewPG SRAM cell is reduced by 31.95% to 45.89% as compared to the conventional 6T SRAM cells depending on the β ratio.

V. CONCLUSIONS

A novel power gated 6T SRAM circuit with multi-threshold voltage transistors is presented in this paper to simultaneously suppress leakage power consumption and enhance data stability as compared to a previously published power gated memory circuit. With the new SRAM circuit, the leakage power consumption is reduced by up to 94.02% and 73.97% as compared to the standard single regular-$|V_{th}|$ and the previously published power gated 6T SRAM circuits, respectively. Furthermore, the read SNM is enhanced by up to 4.79x with the new SRAM circuit as compared to the conventional 6T SRAM circuits. The write power consumption is reduced by up to 13.77% with the new SRAM circuit as compared to the previously published power gated 6T SRAM circuit. A new write assist circuitry is also proposed to enhance the write margin with the new power gating technique. The tradeoffs among different design metrics with the SRAM circuits are summarized in Table IV.

TABLE IV. PERFORMANCE COMPARISON OF DIFFERENT SRAM CIRCUITS

Primary Design Metric	Best Technique	Worst Technique
Read SNM	NewPG	STANDARD ($\beta = 1$)
Leakage Power	NewPG	STANDARD ($\beta = 3$)
Read Power	NewPG (reading "0")	NewPG (reading "1")
Write Power	NewPG	OldPG ($\beta = 3$)
Write Margin	STANDARD/ OldPG ($\beta = 1$)	NewPG

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